METRIC
DOD-C-24529(SH)
AMENDMENT 6
14 January 1988
SUPERSEDING
AMENDMENT 5
6 September 1983

MILITARY SPECIFICATION

CHARGERS, BATTERY, LIFT TRUCK AND
PALLET TRANSPORTER BATTERY SERVICE (METRIC)

This amendment forms a part of Military Specification DOD-C-24529(SH), dated 25 March 1976, and is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1, under "SPECIFICATIONS, MILITARY": Delete reference to "MIL-L-3661" and add "MIL-S-22885 - Switch, Push Button Illuminated, General Specification for."

2.1, under "STANDARDS, MILITARY": Add: "MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of."

PAGE 3

* 3.1: Delete and substitute:

"3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3)."

PAGE 4

3.2.1.3: Delete and substitute:

"3.2.1.3 D.c. output characteristics. Equipment shall operate as specified with variation of input voltage and ambient temperature as specified in 3.2.1.1 and 3.2.2.2, and shall have the following characteristics:

(a) Current adjustment. The charger shall be capable of providing different maximum output currents for different Ah capacity batteries as selected by a four position switch. The current levels for the various Ah capacity batteries shall be as follows:

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Current value (ampere)	Battery Ah capacity
130	600 to 675
150	675 to 750
180	750 to 825
200	825 to 900

Each current setting shall be clearly marked to indicate the current value and related range of battery Ah capacity on the selection switch. The indicated current limit values shall be set and held within 5 percent over the range of operating parameters stated in 3.2.1.1 and 3.2.2.2.

(b) Operation (battery charging). The charger shall automatically regulate the output current to the values of 3.2.1.3(a) until the voltage required to supply this current reaches 2.35 ± 0.02 volts per cell at 25°C, type IV chargers shall have a selector switch for selecting 12 cell or 18 cell batteries. When the above voltage is reached, the charger shall automatically maintain this voltage as the current decreases to prevent excessive gassing of the batteries being charged.

(c) Temperature compensation. The voltage specified in 3.2.1.3(b) shall be automatically compensated for variations in ambient temperature by changing the value of (b) -5.4 to -7.2 millivolts per cell per °C. This may be accomplished by using a temperature sensing element placed at a position within the charger enclosure which is substantially unaffected by the temperature rise within the charger.

(d) Output cable voltage drop compensation. Automatic voltage and current control shall compensate for voltage drop in the output cable (see 3.2.6.10). Compensation shall be accomplished by use of a voltage compounding control matched to the resistance of the output cable and sensitive to the output current or by use of voltage sensing leads from the charger enclosure to the charging connector. The voltage and current values in 3.2.1.3(a) and 3.2.1.3(b) are specified to return batteries to a complete charge condition (see 6.4.3).

(e) The charger output voltage shall be set in accordance with 3.2.1.3(b) but shall have a range of adjustment (via an internal potentiometer which may be mounted on a printed circuit board) between 2.2 to 2.5 volts per cell at 25°C."

PAGE 6

3.2.4, line 2: Delete "class II A," and substitute "class A4,".

Add new paragraph 3.2.4.1:

"3.2.4.1 The test procedure shall be in accordance with procedures CE01 and CE03 as specified in MIL-STD-462."

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PAGE 8

3.2.6.8, line 2: Delete "MIL-L-3661" and substitute "MIL-S-22885."

3.2.6.9.1, item (d), first sentence: Delete and substitute:

"(d) An indicator light in accordance with MIL-L-3661 shall be provided for each fuse which is not usable in an indicator-type fuseholder."

PAGE 9

3.2.6.10.2, first sentence: Delete and substitute: "Cable shall conform to Underwriters Laboratories (UL) Type W, 2 conductor, AWG 2/0 portable power cable."

PAGE 11

3.3.1.6, second sentence: Delete and substitute: "These components shall typically include on-off contactors, meters, indicator lights and voltage selector switch (see 3.2.6.6.2)."

PAGE 13

3.4.3, item (e): Add "(see 3.2.6.6.2)."

3.4.5, line 3: Delete "current measuring system" and substitute "battery charger".

PAGES 16 AND 17

* 4.2, (a): Delete "Qualification" and substitute "First article".

* 4.3 through 4.3.2: Delete and substitute:

"4.3 First article tests. First article tests shall be conducted at a laboratory satisfactory to the Naval Sea Systems Command (NAVSEA). First article tests shall consist of the examination and tests specified in table II and shall be conducted in the order listed.

"4.3.1 Sample for first article tests. One battery charger shall be subjected to the tests specified in table II. Where first article is desired for more than one type charger, acceptance of the highest type classification (see 1.2) will qualify all lower types.

"4.3.2 The supplier shall submit detailed drawings (see 3.6.1) and a test plan to NAVSEA. Manufacture of equipment for tests shall not begin until drawings and test plan are approved by NAVSEA (except for prototype breadboard samples which may be built any time by the supplier to prove certain operating features and circuitry). The test plan shall detail the following:

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(a) Method of performing each test.
(b) Test instrumentation to be used.
(c) The specifications of the test equipment (accuracy, range and frequency response, if applicable).
(d) Diagrams showing the connections to be made to the test instruments.

TABLE II. First article, routine, selective, and periodic inspection.

Description	First article	Routine	Selective	Periodic	Reference paragraph	
					Requirement	Inspection
General examination	X	X	—	X	—	4.5
Mass	X	—	X	X	3.3.1.4	4.6.1
Creepage and clearance distance	X	—	—	X	MIL-E-917	4.6.2
Insulation resistance	X	X	—	X	3.2.5.1.1	4.6.3
Effectiveness of enclosure	X	—	—	X	3.3.1.1	4.6.4
D.c. output and special performance requirements	X	—	X	X	3.2.1.3	4.6.5
Heating	X	—	—	X	3.3.1.5	4.6.6
Phase balance and grounding	X	X	—	X	3.2.1.2	4.6.7
Efficiency	X	—	X	X	3.2.1.9	4.6.8
Inclined operation	X	—	X	—	3.2.2.1	4.6.9
Environmental/ humidity	X	—	—	—	3.2.2.3	4.6.10
Shock	X	—	—	—	3.2.2.1	4.6.11
Vibration	X	—	—	—	3.2.2.1	4.6.12
Voltage withstand	X	X	—	X	3.2.5.1.2	4.6.13
Airborne noise	X	—	—	—	3.2.3.1	4.6.14
Structureborne noise	X	—	—	—	3.2.3.2	4.6.14
Electromagnetic interference	X	—	—	X	3.2.4	4.6.15

"

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PAGES 19 AND 20

* 4.6, first sentence: Delete "qualification" and substitue "first article".

4.6.1: Delete and substitute:

"4.6.1 Size and mass. The size and mass of the battery charger and the size and mass of the onboard repair parts shall be taken and recorded separately. The size and mass of the battery charger shall not exceed the value specified in 3.3.1.4."

4.6.4, make the following changes: Line 2, delete: "corresponding type of enclosure except for". Line 3, after "shall" add: "also".

4.6.5.1: Delete and substitute:

"4.6.5.1 Output characteristics selective and periodic tests only. Battery chargers shall be tested under all conditions of voltage, frequency, ambient temperature and load within specified rating range. The output characteristics (see 3.2.1.3) shall remain within the specified limit without adjustment other than initial setting. Output characteristic tests shall be conducted as follows:

(a) Temperature 25 ± 2°C.
(b) Resistive load cycle - 200 amperes, 440 V, 60 Hz for 1 hour
100 amperes, 462 V, 60 Hz for 1 hour
15 amperes, 418 V, 60 Hz for 1 hour
200 amperes, 462 V, 60 Hz for 1 hour

Each load step shall be applied immediately after previous load condition.

(c) Initial output voltage setting - nominal.
(d) Checks shall be made to see that compensation is automatically made by the charger regulation for ambient temperature fluctuation affecting charge voltage.
(e) Verification shall be made of the automatic compensation for voltage drop between charger and battery based on a cable length of 4.25m of 2-conductor (1.28cm^2 each) load cable.
(f) Input voltage, frequency and current, output current, and voltage shall be recorded every 30 minutes."

PAGE 20

4.6.5.2, make the following changes: Lines 2 and 3, delete: "or by connection of transformer taps in a specific manner". Lines 5 and 6, delete: "or of connecting transformer leads or taps".

PAGE 22

4.6.6, line 2: Delete "4.6.1.1" and substitute "4.6.6.1.1".

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* 4.6.11.1(f)(1), line 2: Delete "qualification" and substitute "first article".

PAGE 25

4.7.1, line 2: After "with" add: "the requirements of 3.3.1.7.2(d)(3) and".

PAGES 26 and 27

6.1.2: Delete and substitute:

"6.1.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.6.1	Drawings, engineering and associated lists	DI-E-7031	Level 3 Design activity designation - contractor Design activity drawing numbers - contractor Parts list and data list - DOD-STD-100 Hard copy of drawings - required (see 4.3.2) Certification data sheets - required
3.6.1	Imaged aperture/ tabulating cards	DI-E-20477	Microfilm of drawings - type I, class 1 delivered to: Commander Portsmouth Naval Shipyard Naval Engineering Drawing Support Activity, Code 202.2 Portsmouth, NH 03801

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.6.2	Manual, technical, standard	DI-M-2044	MIL-M-15071, type II
3.6.3.2	System safety engineering report	DI-H-7050	----
3.6.4	Provisioning parts list (PPL)	DI-V-7002	----
4.1.1	Inspection system program plan	DI-R-4803	----
4.7	Report, test/inspection	UDI-T-23473	----
4.7	Report, failure/malfunction	UDI-T-23724	----
4.7	Procedures, test	UDI-T-23732	----
4.7.1	Reliability predictions report	DI-R-7082	----
4.7.1	Reliability prediction and documentation of supporting data	DI-R-7095	----
4.7.2	Plan, maintainability demonstration	DI-R-2129	----
4.7.3	Report, maintainability demonstration	DI-R-7113	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)"

PAGE 27

* 6.3: Delete and substitute:

"6.3 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract."

* 6.3.1: Delete.

Add as paragraph 6.4.8:

"6.4.8 Battery gassing voltage. The battery gassing voltage for lead acid batteries is described in NAVSEA S9086-KR-STM-000/CH313."

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Add as new paragraph 6.7:

"6.7 Typical operation. It is the intent of the current limit values associated with the various ampere-hour capacity batteries to provide the maximum charging current feasible to charge the battery in the least amount of time, while keeping the battery temperature rise above ambient less than 9°C. The end of charge current to the batteries will typically be less than 1/60 of the ampere-hour rating of the batteries, so that no detrimental over-charging of the battery occurs if the battery remains connected to the charger for 24 hours after the charging current has decreased to 1/60 of the battery ampere-hour rating."

LAST PAGE

DD 1426, Standardization Document Improvement Proposal: Delete address and substitute:

"COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362-5101"

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Preparing activity:
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